

Calculus 1 exam (20 January 2025)
Duration: 1h30min

Exercise 1 (6points)

A) Let z be a complex number, we set $P(z) = z^3 + iz^2 + iz + 1 + i$

1. Calculate $P(-1 - i)$.
2. Determine a and b such that for all $z \in \mathbb{C}$:

$$P(z) = (z + 1 + i)(z^2 + az + b).$$

3. Solve in $\mathbb{C}$ the equation $P(z) = 0$.

B) Let the complex number $z = 1 + i\sqrt{3}$.

1. Write z in the trigonometric form.
2. Find the integer $n \in \mathbb{N}$ such that $(1 + i\sqrt{3})^n$ is a positive real number.

Exercise 2 (7points)

A) Calculate the following limits:

$$\lim_{n \rightarrow +\infty} \frac{2^{n+1} + 3^{n+1}}{2^n + 3^n}, \quad \lim_{n \rightarrow +\infty} \frac{\sin n}{n}.$$

B) Let consider the recurring sequence $(U_n)_n$ defined by:

$$\begin{cases} U_1 &= \sqrt{2} \\ U_{n+1} &= \sqrt{2 + U_n}, \quad \forall n \geq 1. \end{cases}$$

1. Prove that for all $n \in \mathbb{N}^*$, $U_n \leq 2$.
2. Prove that for all $n \in \mathbb{N}^*$, $2 - U_{n+1} < \frac{2 - U_n}{2}$.
3. Deduce the nature of $(U_n)_n$ and calculate its limit if it exists.
4. Let $A = \{U_n, n \in \mathbb{N}^*\}$. Determine $\sup A$, $\inf A$, $\max A$ and $\min A$ if they exist.

Exercise 3 (7points) Let f be the function defined by

$$f(x) = \ln(x + \sqrt{x^2 - 1}).$$

1. Determine the domain of definition D_f of f .
2. Study the continuity of f on D_f .
3. Study the differentiability of f on $]1, +\infty[$, and calculate its derivative f' .
4. Prove that f admits an inverse function denoted by f^{-1} and precise its domain of definition with its arrival domain.
5. We set $x_0 = 2$, $y_0 = f(x_0) = \ln(2 + \sqrt{3})$. Calculate $(f^{-1})'(y_0)$.
6. Using the mean value theorem, prove that for all $x > 1$,

$$\frac{\ln(x + \sqrt{x^2 - 1})}{x - 1} > \frac{1}{\sqrt{x^2 - 1}}.$$

and deduce $\lim_{x \rightarrow 1^+} \frac{\ln(x + \sqrt{x^2 - 1})}{x - 1}.$

Remark: Exercises 2 and 3 are considered as a written interrogation.